

Fundamentals Before Hibernate

Persistence:

- **Definition:** The process of storing and managing data for a long time is called persistence.
- **Examples:**
 1. **File Operations:** Storing data in files on a hard disk (e.g., using Java I/O operations).
 2. **Database Storage:** Storing data in the form of tables in a database (e.g., using SQL).

Persistence Store:

- **Definition:** A persistence store is a place or storage where data is saved and managed for an extended period.

Persistent Data:

- **Definition:** The data stored in a persistence store is referred to as persistence data.

Persistence Operations:

- **Definition:** The operations performed on persistent data, such as creating, reading, updating, or deleting data.

Persistence Logic:

- **Definition:** The logic written to perform persistence operations is called persistence logic. This logic ensures data is correctly stored and retrieved.

Persistence Technology/Framework:

- **Definition:** The technology or framework used to write persistence logic is known as persistence technology or framework.
- **examples:**
 - **JDBC (Java Database Connectivity):** A technology for connecting and executing queries against the database.

- **Hibernate:** An ORM (Object-Relational Mapping) framework that simplifies database interactions by mapping Java objects to database tables.
- **Spring JDBC / Spring ORM / Spring Data JPA:** Frameworks provided by Spring to manage persistence logic with advanced features and integration.

Additional Notes:

- ***Persistence in Modern Applications:*** Modern applications often require persistent data storage to ensure that data is not lost when the application is closed or the system is restarted. Technologies like Hibernate abstract the complexity of database interactions, allowing developers to focus on business logic.
- ***Importance of ORM:*** ORM frameworks like Hibernate automate many of the tedious aspects of database management, such as writing SQL queries and managing connections, which can lead to faster development and more maintainable code.